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Hippo: conjugate heat transfer

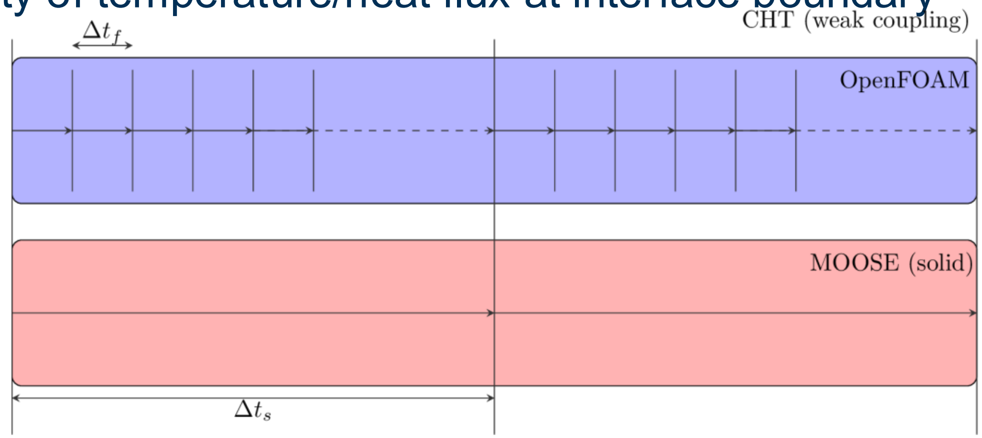
Matthew Falcone

Overview

- Weakly coupled CHT
- Coupling strategies
  - Temperature-forward flux back (TFFB)
  - Flux-forward temperature back (FFTB)
  - Heat transfer coefficient-forward flux back (hFFB)
  - Heat transfer coefficient-forward flux back (hFTB)
- Stability of the coupling strategies
- Exercises
  - Flow over a heated plate
    - TFFB and FFTB
  - Shell and tube heat exchanger
    - hFTB

Weakly-coupled CHT

- Where the fluid and solid thermal fields and solved sequentially.
  - At the end of the step exchange temperature/heat flux.
  - Alternatives include:
    - Solving thermal fields of fluid and solid in the same matrix
    - Use fixed-point iteration to converge boundary temperature/heat flux
- Main advantage is computational cost
  - The fluid field usually requires much smaller time steps than the solid
    - We can take advantage of sub-cycling
- Main disadvantage:
  - Lack of continuity of temperature/heat flux at interface boundary



Coupling strategies for CHT

- Temperature-Forward Flux-Back (TFFB)
  - Temperature sent from fluid to solid
  - Heat flux sent from solid to fluid
- Flux-Forward Temperature Back (FFTB)
  - Heat flux sent from fluid to solid
  - Temperature sent from solid to fluid
- Heat-transfer coefficient Forward Temperature Back (hFTB)
  - HTC and bulk temperature sent from fluid to solid
  - Heat flux sent from solid to fluid
- Heat-transfer coefficient Forward Flux Back (hFFB)
  - HTC and bulk temperature sent from fluid to solid
  - Temperature sent from solid to fluid

Stability of CHT coupling strategies

- The stability of the coupling depends on the conditions on each side of the interface.
  - Verstraete and Scholl (2016) highlighted the importance of the Biot number

$$Bi = h\,L_s/k_s$$

    - FFTB and hFTB are stable for Bi < 1 and TFFB and hFFB are stable for Bi > 1, although with many caveats
  - The analysis doesn't account for transient effects which at some point I want to consider
    - Density and heat capacity of the fluid and solid are also important for stability
- The basic instability mechanism is due to growing heat flux oscillations on the interface
  - Large temperature changes on a time step leads to growing reversals of heat flux

TFFB in Hippo

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- Some key parts from theflow over heated plate TFFB example
  - AuxKernel to calculate heat flux on solid-side
  - FoamDiffusionFluxBC imposes it
  - Temperature copied with FoamVariableField
  - MatchedValueBC imposes it

```
[AuxVariables]
    [fluid_wall_temperature]
        family = LAGRANGE
        order = FIRST
        initial_condition = 0
    []
    [wall_heat_flux]
        family = MONOMIAL
        order = CONSTANT
        initial_condition = 0
    []
[]

[AuxKernels]
    [heat_flux_aux]
        type = DiffusionFluxAux
        variable = wall_heat_flux
        diffusion_variable = T
        boundary = 'solid_top'
        diffusivity = thermal_conductivity
        component = normal
    []
[]

[BCs]
    [fixed_temp]
        type = DirichletBC
        variable = T
        boundary = solid_bottom
        value = 310
    []

    [fluid_interface]
        type = MatchedValueBC
        variable = T
        boundary = solid_top
        v = fluid_wall_temperature
    []
[]
```

```
[Transfers]
    [wall_temperature_from_fluid]
        type = MultiAppGeometricInterpolationTransfer
        source_variable = fluid_wall_temp
        from_multi_app = hippo
        variable = fluid_wall_temperature
        execute_on = same_as_multiapp
    []

    [heat_flux_to_fluid]
        type = MultiAppGeometricInterpolationTransfer
        source_variable = wall_heat_flux
        to_multi_app = hippo
        variable = solid_heat_flux
        execute_on = same_as_multiapp
    []
[]
```

```
[FoamBCs]
    [solid_heat_flux]
        type = FoamDiffusionFluxBC
        foam_variable = T
        initial_condition = 0
    []
[]

[FoamVariables]
    [fluid_wall_temp]
        type = FoamVariableField
        foam_variable = T
        initial_condition = 300
    []
[]
```

FFTB in Hippo

- Some key parts from theflowover heated plate FFTB example
  - ProjectionAux to create constant monomial
  - FoamFixedValueBC imposes it
  - Heat flux calculated with OpenFOAM function object
  - CoupledVarNeumannBC imposes it

```
[AuxVariables]
    [solid_wall_temperature]
        family = LAGRANGE
        order = FIRST
        initial_condition = 300
    []
    [fluid_heat_flux]
        family = MONOMIAL
        order = CONSTANT
        initial_condition = 0
    []
[]

[AuxKernels]
    [projection_aux]
        type = ProjectionAux
        variable = solid_wall_temperature
        v = T
        boundary = 'solid_top'
    []
[]

[BCs]
    [fixed_temp]
        type = DirichletBC
        variable = T
        boundary = solid_bottom
        value = 310
    []

    [fluid_interface]
        type = CoupledVarNeumannBC
        variable = T
        boundary = solid_top
        v = fluid_heat_flux
        scale_factor = -1.
    []
[]
```

```
[MultiApps]
    [hippo]
        type = TransientMultiApp
        app_type = hippoApp
        execute_on = timestep_begin
        input_files = 'fluid.i'
    []
[]

[Transfers]
    [heat_flux_from_fluid]
        type = MultiAppGeometricInterpolationTransfer
        source_variable = fluid_heat_flux
        from_multi_app = hippo
        variable = fluid_heat_flux
        execute_on = same_as_multiapp
    []

    [temp_to_fluid]
        type = MultiAppGeometricInterpolationTransfer
        source_variable = solid_wall_temperature
        to_multi_app = hippo
        variable = solid_wall_temp
        execute_on = same_as_multiapp
    []
[]
```

```
[FoamBCs]
    [solid_wall_temp]
        type = FoamFixedValueBC
        foam_variable = T
        initial_condition = 300
    []
[]

[FoamVariables]
    [fluid_heat_flux]
        type = FoamFunctionObject
        foam_variable = "wallHeatFlux"
        initial_condition = 0
    []
[]
```

hFTB in Hippo

- Why might we choose to use hFTB and not FFTB
  - Verstraete and Scholl (2016) suggests stability might be similar
- In some case, we only care about the steady state solution
  - At each coupling step, we can do a steady solve – Poisson equation
    - Generally, much faster convergence
  - However, the Laplace operator is singular with all-Neumann BCs
    - Frequently happens using FFTB and *usually* can't be solved
    - Therefore, it must be run transient.
  - hFTB can be run steady-state
- To use the HTC, we use a special FoamVariable called FoamHeatTransferCoeff
  - It also requires a method of calculating the bulk temperature.
  - In Hippo, these are implemented as UserObjects
    - AdjacentCellBulkTemperature
    - FixedBulkTemperature

hFTB in Hippo

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- Note user object and FoamVariable
- CoupledConvectiveHeatFluxBC used to impose the HTC boundary condition
- Bulk temperature UO and the HTC needs to be transferred

```
[AuxVariables]
    [inner_htc]
        family = MONOMIAL
        order = CONSTANT
        initial_condition = 0
    []
    [bulk_t_inner]
        family = MONOMIAL
        order = CONSTANT
    []
```

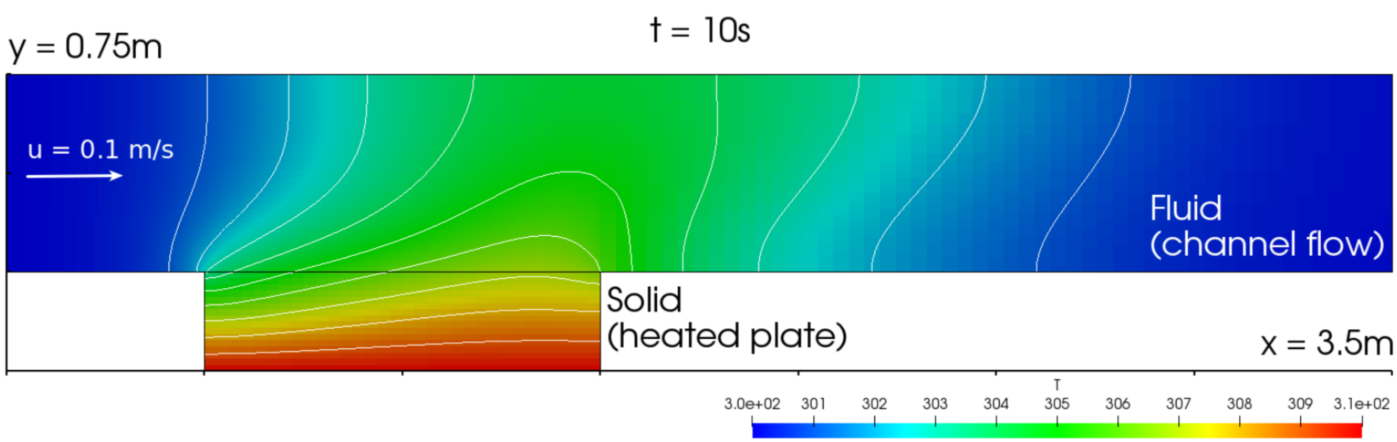
```
[FoamVariables]
    [fluid_htc]
        type = FoamHeatTransferCoeff
        bulk_temperature_uo = bulk_t
        boundary = interface
    []
[]

[UserObjects]
    [bulk_t]
        type = AdjacentCellBulkTemperature
        boundary = interface
        execute_on = 'initial timestep_end'
    []
[]
```

```
[inner]
    type = CoupledConvectiveHeatFluxBC
    variable = T
    boundary = inner
    T_infinity = bulk_t_inner
    htc = inner_htc
[]
```

```
[htc_from_outer]
    type = MultiAppGeneralFieldNearestLocationTransfer
    source_variable = fluid_htc
    from_multi_app = outer
    variable = outer_htc
    execute_on = same_as_multiapp
    to_boundaries = 'outer'
    search_value_conflicts=false
[]
[bulk_from_inner]
    type = MultiAppGeneralFieldUserObjectTransfer
    source_user_object = bulk_t
    from_multi_app = inner
    variable = bulk_t_inner
#    bbox_factor = 2.
    to_boundaries = 'inner'
    greedy_search = false
[]
```

Example 1: Flow over a heated plate

- Derives from preCICE example
- Transient heat conduction in the heated plate modelled using MOOSE
  - Using TFFB and FFTB
- For FFTB, we will consider different physical properties highlighting the stability of the CHT schemes



Example 2: Shell and tube heat exchanger

- Also derives from a preCICE tutorial
- Two OpenFOAM simulations
  - Tube (inner)
  - Shell (outer)

- MOOSE used for solid heat conduction
  - Replaces CalculiX in the tutorial
  - Both use the same meshes as the preCICE example
- Uses hFTB with solid run steady-state